RARE DISEASE, REAL KID · MVA HACKATHON 2026 · TRACK 2 REPORT

Allele-aware drug repurposing for *BUB1B* mosaic variegated aneuploidy

Three evidence streams, a patient-aware safety model, and a ranking that let the evidence change our minds

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Code: github.com/aperezcastro/hackaton-genetico, scripts/t2/ (CC BY 4.0)

All proposals are hypotheses to be tested, not treatment advice. Every decision about this child's care belongs to their clinical team. Only gene-, pathway- and drug-level public data were used.

Recommendations at a glance

Tier 0 · act now	**Safety note** for the clinical team: drugs to discuss and monitor (never to drop a curative regimen), growth hormone, NAD^+ supplements, HSP90 inhibitors; paediatric label items for every candidate	MVA case reports + mouse heart data + connectivity + FDA labels; its own risk is under-treatment, which is why it advises discussion, not substitution
Tier 0 · measure	**BUBR1 immunoblot** in the child's cells, and **%MN-RET** in blood	the blot decides the allele class and so whether a corrective drug can exist; %MN-RET is the first measure of this disease's activity in blood (never made in MVA), then a recent error rate on a weeks timescale
Tier 1 · experiment	**Intermittent dasatinib**, to be tested first in the published BubR1 allelic mouse	highest-scoring approved drug (narrowly ahead of ruxolitinib), and the only top candidate whose direction is tumour-suppressive; genetic senolysis delayed MVA-like phenotypes in BubR1 mice; no senolytic drug tested in that model
Tier 2 · gated	**An NAD^+ precursor** (niacin; nicotinamide scores the same but inhibits sirtuins) to stabilise the hypomorphic protein	becomes the top-ranked option if the blot confirms the low-abundance class *and* the child is in documented remission (rhabdomyosarcoma depends on NAMPT/NAD^+)
Tier 3 · conditioned	**Low-dose sirolimus** for muscle loss, **with %MN-RET as a safety readout**	mTORC1–sarcopenia link is from monoallelic mice; rapamycin was aneugenic in one in-vitro study; adding temsirolimus to chemotherapy raised myelotoxicity in paediatric RMS
Tier 4 · biomarker-gated	**JAK inhibitors**, only if an interferon signature is measured in the child	class-consistent reversal of the knockout signature, but not a top hit; may impair immune clearance of aneuploid cells
Declined	**Proteostasis amplifiers** (arimoclomol, sodium phenylbutyrate) and antioxidants	the lever is real, but raising HSF1 is a documented aneuploidy-*tolerating* event: it makes chromosomally abnormal cells viable
Closed (for now)	Read-through of the nonsense allele	on current evidence: UGA-A context (a two-feature reading, not the published predictor) + NMD; ataluren discredited; the compounds that correct UGA are unapproved, and gentamicin reads through weakly and is nephrotoxic

2 facts (allele class, remission) would move a corrective drug from #8 to #1

0 registered clinical trials for MVA (ClinicalTrials.gov)

32 BUB1B-knockout signatures analysed (20 cell lines)

132 MVA publications screened for treatments

1 Summary

The child carries a **null** *BUB1B* allele (p.Leu737*, UGA nonsense, NMD predicted) and a kinase-domain missense allele (p.Asn1002Lys), as established in Track 1. The mechanism is **partial loss of function**: the SAC is dosage-sensitive (Bohers et al. 2008), and more BUBR1 protects against aneuploidy and cancer in vivo (Baker et al. 2013), so modest gains in functional BUBR1 should help and drugs that add mitotic or

genotoxic stress should be avoided (Figure 1).

What we do not know, and what it costs. Suijkerbuijk et al. (2010) separate MVA alleles into two classes: those that lower BUBR1 abundance and those that impair its function at normal abundance. Which class p.Asn1002Lys belongs to has never been measured. The prior favours low abundance — every disease-associated substitution tested in or near the kinase domain was 5–10× less abundant, and the known stable-but-dysfunctional alleles lie outside that domain — so we weight it 0.75, not 1. We carry both scenarios explicitly (§2.1) rather than assert the convenient one: the corrective NAD^+ line loses a quarter of its mechanistic fit, and one immunoblot becomes the gate for it.

We combined three independent evidence streams in an open, reusable pipeline (Figure 2):

1. **Literature.** A curated, PMID-cited knowledge base of seven therapeutic lines plus a safety card, in which contrary evidence is scored and counts against us.
2. **Connectivity.** The consensus transcriptional signature of *BUB1B* knockout (32 CRISPR signatures, 20 cell lines, LINCS L1000), queried against the LINCS chemical-perturbation signatures (the 1,000 most significant hits per direction).
3. **Patient-aware safety.** FDA label text mined automatically and matched against the child's own phenotype (cancer predisposition, nephrocalcinosis, muscle atrophy, short stature), plus curated literature risks that no label records.

The mechanism-matched candidate exists, and we name it first: an NAD^+ precursor (niacin), which is predicted to act on the defect itself by keeping the hypomorphic BUBR1 from being degraded (H1). We do not rank it first today, for two reasons that are facts about this child and can be measured, not gaps in the literature: whether the missense allele is of the low-abundance class (one immunoblot), and whether the child is in remission (their tumour depends on NAD^+).

The main result is a decision rule, not a single drug (§4.2). We classify every line by what it does to a chromosomally unstable cell — **correct** fewer missegregation events, **clear** the abnormal cells, **tolerate** help them cope. Under today's uncertainty the best-supported approved option, **intermittent dasatinib**, is on the *clear* axis: it ranks first, narrowly ahead of the JAK inhibitor ruxolitinib (0.73 vs 0.72; first in 42 % and 38 % of all robustness draws), and it is the only top candidate whose direction is tumour-suppressive. The corrective NAD^+ line ranks eighth, held back by two named facts about this child: the unmeasured allele class and the rhabdomyosarcoma's dependence on NAD^+. **If the clinical team establishes either one, an NAD^+ precursor rises to third; with both, it ranks first.** The two *tolerate* lines (mTOR and JAK inhibitors) share a different concern: both can blunt the immune clearance of aneuploid cells, and in a cancer-predisposition syndrome a premalignant clone is an aneuploid cell too.

The second result is how much the evidence **changed our own plan** (Figure 5).

- Our initial lead, nicotinamide, fell from second to ninth (niacin, now the line's preferred precursor, is eighth): rhabdomyosarcoma, this child's tumour, depends on the NAD^+ pathway it would feed, and nicotinamide itself inhibits sirtuins.
- The read-through line was closed, on current evidence, by an allele-level analysis of the stop codon.
- The best human senolytic trial missed its primary endpoint, so senolysis became an experiment to run rather than a treatment to propose.
- Sirolimus, once our top-scoring drug, fell to tenth: the mTORC1–sarcopenia link it rested on comes from monoallelic mice, rapamycin was aneugenic in one in-vitro study (Bonatti et al. 1998), and adding the mTOR inhibitor temsirolimus to chemotherapy raised myelotoxicity in paediatric RMS (Gupta et al. 2024).
- A commonly proposed alternative for this genotype — proteostasis amplification — was added as a line and then declined on the evidence, rather than left unmentioned (§4.4).
- A blind adversarial review of an earlier draft showed that our previous headline depended on two hand-set inputs and on harms counted twice (once against the evidence, once against safety). We separated the two in the model, widened the robustness analysis, and replaced that headline with the decision rule above.

Impact beyond this child. The safety card, the screen of every published MVA case and the knockout signature are not specific to these two alleles; they apply to anyone with MVA1. We also propose a blood

measurement — micronucleated reticulocytes (%MN-RET) — that has **never been made in anyone with MVA**. Its first use is not about drugs: it would be the first quantitative readout of this disease's activity in blood, from a routine draw, against a published paediatric reference range. That is a diagnostic and natural-history observation for every MVA patient, and it can falsify the disease model before anyone is treated. Only then does it become an endpoint that reports a **recent error rate on a weeks timescale**, with a one-page protocol and the power calculation that says what it can and cannot resolve (§5). **Innovation:** the therapeutic-axis model and the decision rule it yields; a *negative* output (what not to give); safety that goes beyond drug labels; an evidence model that scores target validation below demonstrated benefit; and an allele-class uncertainty carried explicitly into the score instead of being resolved by assumption.

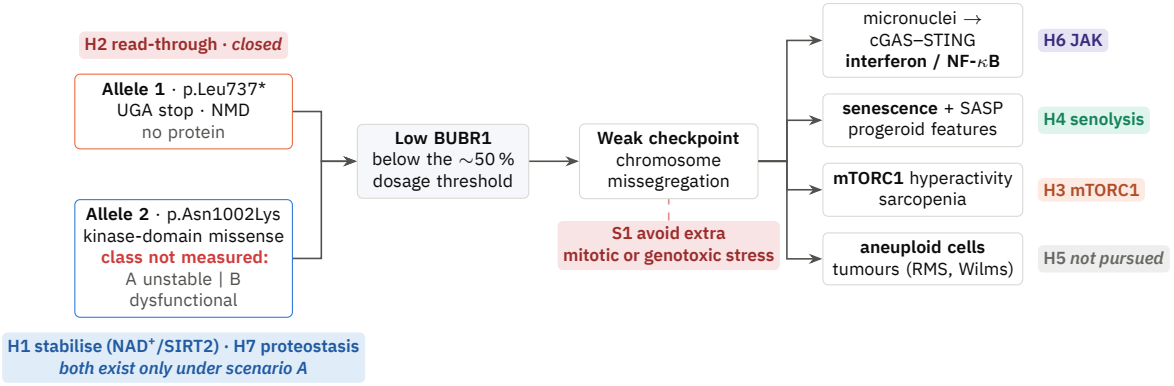


Figure 1. Disease mechanism and where each therapeutic line acts. Each allele has its own lever (H1 and H7 on the missense allele, H2 on the nonsense one), and the two missense-allele levers are conditional on an allele class nobody has measured (§2.1); the other lines act on druggable consequences of chronic missegregation; the safety card (S1) acts on the weak checkpoint itself. Colour identifies the line, not the axis; for the axes see §4.2.

2 Disease mechanism model

Table 1 summarises the mechanism, level by level, with its evidence.

2.1 The one fact that would change everything, and we do not have it

Two scenarios for p.Asn1002Lys, carried in parallel

	A · low abundance	B · stable, dysfunctional
Precedents in the same study	I909T, L1012P, R727C, R814H, L844F: 5–10× less protein, turnover doubled, HSP90- and proteasome-dependent	Y155C (stable; restores alignment but not the checkpoint) and R550Q (stable) — both *outside* the kinase domain. The one stable kinase-domain substitution, Q921H, is a functionally normal passenger in *cis* with the causal L844F
What the defect is	scarcity	the molecule itself
Consequence for therapy	abundance is the lever: forced re-expression of I909T and L1012P *fully* restored the checkpoint, so there is nothing wrong with the protein except that there is too little of it. H1 is on-mechanism	stabilising it achieves nothing, and no approved drug restores the function. The actionable set collapses to the safety card and the downstream lines
Prior we assign	0.75	0.25

The kinase-domain position makes scenario A likely but not certain: this residue has never been characterised, and the child's survival, where the mouse X753/L1002P died, might mean that p.Asn1002Lys keeps more function than L1012P. Three consequences follow.

1. **The uncertainty is priced into the mechanistic fit, not into the evidence.** The studies behind H1 are exactly as strong as they were; what is uncertain is whether *this* child is the kind of patient they apply to. So H1 gets $M = 0.8 \times 0.75 = 0.6$, while E is left untouched. Putting it in the right term matters: it moves the NAD⁺ line down the ranking without pretending the underlying biology is weaker than it is.

Level	Fact	Evidence
Gene	*BUB1B* encodes BUBR1, a core subunit of the mitotic checkpoint complex that inhibits APC/C–CDC20 until all kinetochores are attached; it also stabilises kinetochore–microtubule attachments	Baker et al. 2013, PMID 23242215
Allele 1	nonsense p.Leu737*, exon 17/23, NMD; truncating MVA alleles yield no transcript	Suijkerbuijk et al. 2010, PMID 20516114
Allele 2	kinase-domain missense. **Class not measured** (§2.1): low-abundance (prior 0.75: every disease-associated substitution tested in or near the kinase domain was 5–10× less abundant) or stable-but-dysfunctional (0.25: known examples lie outside the kinase domain)	Suijkerbuijk et al. 2010
Structure	Asn1002 is a high-confidence (AlphaFold pLDDT 91), largely buried residue packing against the kinase C-lobe, 12 Å from the MVA1 allele L1012P; a longer, charged Lys could destabilise the fold (consistent with scenario A, not proof of it)	`scripts/t2/estructura.py`
Dosage	aneuploidy appears below ∼50 % residual BUBR1; graded effects across 8–77 %	Bohers et al. 2008, PMID 18932004
Rescue logic	ectopic BUBR1 restores the checkpoint in MVA patient cells; transgenic BubR1 overexpression reduces aneuploidy and cancer in mice	Suijkerbuijk et al. 2010; Baker et al. 2013
Cellular response	knockout: interferon-α/γ, NF-κB and p53 programmes up, proliferation down (§4.3), consistent with micronuclei sensed by cGAS–STING	Mackenzie et al. 2017, PMID 28738408; Santaguida et al. 2017, PMID 28633018
Phenotype drivers	aneuploidy leads to tumours, senescence + SASP (sarcopenia, lipodystrophy, cataract) and growth failure	Baker et al. 2004; Baker et al. 2011; Sieben et al. 2020
Genotype-specific	in a BubR1 allelic series, the truncating + kinase-domain missense genotype closest to this child (X753/L1002P; mouse L1002 = human L1012) **died prematurely**; the viable biallelic model (H/L1002P) shows the MVA spectrum; sarcopenia correlated with mTORC1 hyperactivity in mice carrying a **single** MVA allele	Sieben et al. 2020, PMID 31738183
Feedback loop	**a prediction specific to this genotype.** Kinase-domain BUBR1 mutants need HSP90 to fold and are cleared by the proteasome without it; and aneuploid human cells are *already* impaired in HSP90-dependent folding, with a compromised heat-shock response. If allele 2 is of the low-abundance class, aneuploidy erodes the chaperone reserve that the hypomorphic allele depends on, in the very cells that need it most: less BUBR1 → more aneuploidy → worse folding → less folded BUBR1	Suijkerbuijk et al. 2010, PMID 20516114; Donnelly et al. 2014, PMID 25205676

Table 1. Disease mechanism model, level by level, with the evidence for each step.

2. **One experiment decides it**, and it is cheap: a BUBR1 immunoblot on the child's cells, with a cycloheximide chase if the level is low. That is rung 1 of the validation ladder (§6), and it is the gate for every restorative proposal here.
3. **The decision rule is pre-declared.** Residual full-length BUBR1 clearly below the parental carrier range, with accelerated turnover, supports scenario A and opens H1 and H7. A level in the carrier range with normal turnover supports scenario B and **closes** both, leaving tiers 0, 1, 3 and 4 — none of which depends on the allele class.

3 Methods: the pipeline

Score components.

- **E, evidence that the line works** (0–1): weight of the strongest supporting study, +0.1 per additional one (max +0.2), minus $0.3\times$ the weight of each contrary study (max -0.3), so that E does not depend on how hard we searched for contrary evidence. Evidence of **benefit** weighs human same gene 1.0, mouse same gene 0.8, human other 0.7, cell same gene 0.5, mouse other 0.5, non-mammalian animal 0.4, cell other 0.3. **Target validation** (the pathway is disturbed, no benefit of modulating it shown) weighs 0.5 or 0.3, and **feasibility** in another disease (tolerated, no benefit for this mechanism) weighs 0.3. A pathway being active is not evidence that blocking it helps.
- **M, mechanistic fit** to *this* genotype (0–1, curated and justified in the YAML).
- **P, paediatric feasibility**: label with paediatric use or dosing 1.0; label without 0.6; approved without a US label 0.5; supplement 0.35.

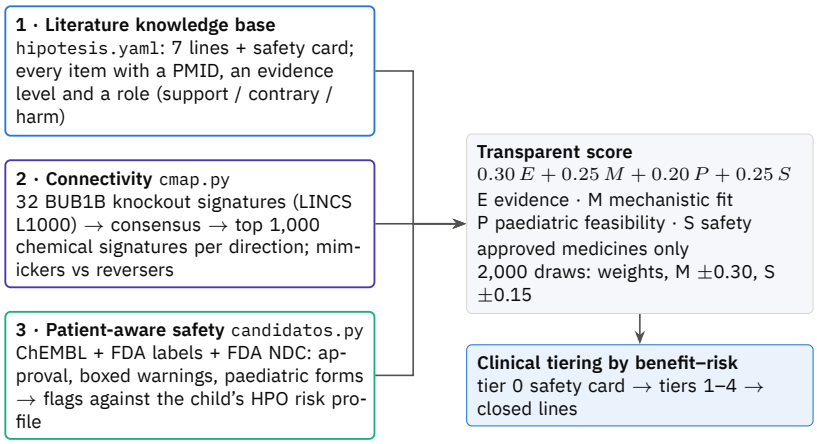


Figure 2. Three independent evidence streams feed one transparent score; the clinical tiering is then argued explicitly. The connectivity stream is also used to challenge our own candidates, not only to find new ones.

- **S, patient-specific safety**: $1 - 0.25$ per boxed-warning flag $- 0.1$ per warning flag, minus a **curated literature risk** for hazards no label records (max penalty 0.7). Evidence that a line could *harm this child* lowers S only; it no longer also lowers E. The curated risks: the NAMPT dependency of rhabdomyosarcoma (H1, 0.35), aneuploidy tolerance (H7, 0.3), aneugenicity and paediatric myelotoxicity (H3, 0.2), weakened immune clearance of aneuploid cells (H6, 0.2).

Safety weighs as much as mechanism because this is a child with cancer predisposition. Connectivity is shown next to the score but not inside it, because the signatures come from tumour cell lines after acute knockout.

4 Results

4.1 Ranked candidates

Only **approved medicines** are ranked, matching the Track 2 scope confirmed by the organisers (discussion #5). Quercetin, fisetin and nicotinamide riboside are dietary supplements and are listed but not ranked; ataluren was removed on efficacy grounds (§4.6).

#	Drug	Line	Axis	E	M	P	S	Score	Paediatric form	Key patient-specific flag
1	dasatinib	H4	clear	0.6	0.60	1.0	0.80	**0.73**	oral suspension	pleural effusion, PAH, paediatric bone growth (label)
2	ruxolitinib	H6	tolerate	0.9	0.40	1.0	0.60	0.72	solid only	malignancy (oral label, no boxed warning); may weaken immune clearance
3	tofacitinib	H6	tolerate	0.9	0.40	1.0	0.45	0.68	oral solution	malignancy (boxed); may weaken immune clearance
4	fedratinib	H6	tolerate	0.9	0.40	0.6	0.70	0.67	solid only	malignancy; boxed encephalopathy (Wernicke; not a patient-risk flag, so not scored)
5	arimoclomol	H7	correct	0.7	0.30	1.0	0.70	0.66	solid only	**declined**: aneuploidy tolerance (§4.4)
6	acetylcysteine	H7	correct	0.7	0.30	1.0	0.70	0.66	oral/IV solution	**declined**: aneuploidy tolerance
7	sodium phenylbutyrate	H7	correct	0.7	0.30	1.0	0.70	0.66	pellets	**declined**: aneuploidy tolerance
8	**niacin**	H1	correct	0.85	0.60	0.6	0.45	0.64	oral solution	**NAMPT dependency of RMS**; label: myopathy, renal
9	nicotinamide	H1	correct	0.85	0.40	0.6	0.65	0.64	solid only	NAMPT dependency of RMS; inhibits sirtuins
10	sirolimus	H3	tolerate	0.55	0.60	1.0	0.45	0.63	oral solution	aneugenic in vitro; malignancy (boxed)
11	baricitinib	H6	tolerate	0.9	0.40	0.6	0.45	0.60	solid only	malignancy (boxed)
12	everolimus	H3	tolerate	0.55	0.60	1.0	0.30	0.59	dispersible tablet	malignancy, nephrotoxicity (boxed)
13	hydroxychloro-quine	H5	clear	0.3	0.40	1.0	0.70	0.57	solid only	cell-line evidence only
14	metformin	H5	clear	0.3	0.40	1.0	0.65	0.55	oral solution	cell-line evidence only
15	amlexanox	H2	correct	0.4	0.15	0.5	1.00	0.51	solid only	UGA-A context + NMD (§4.6)

Table 2. Ranked approved medicines (full table with every column: `reports/t2/candidatos.md`). The *axis* column is the result that matters: see §4.2. Ties (5–7, 8–9) are listed in the knowledge base's order. Nicotinamide is sold mostly as an over-the-counter vitamin; niacin is a prescription medicine.

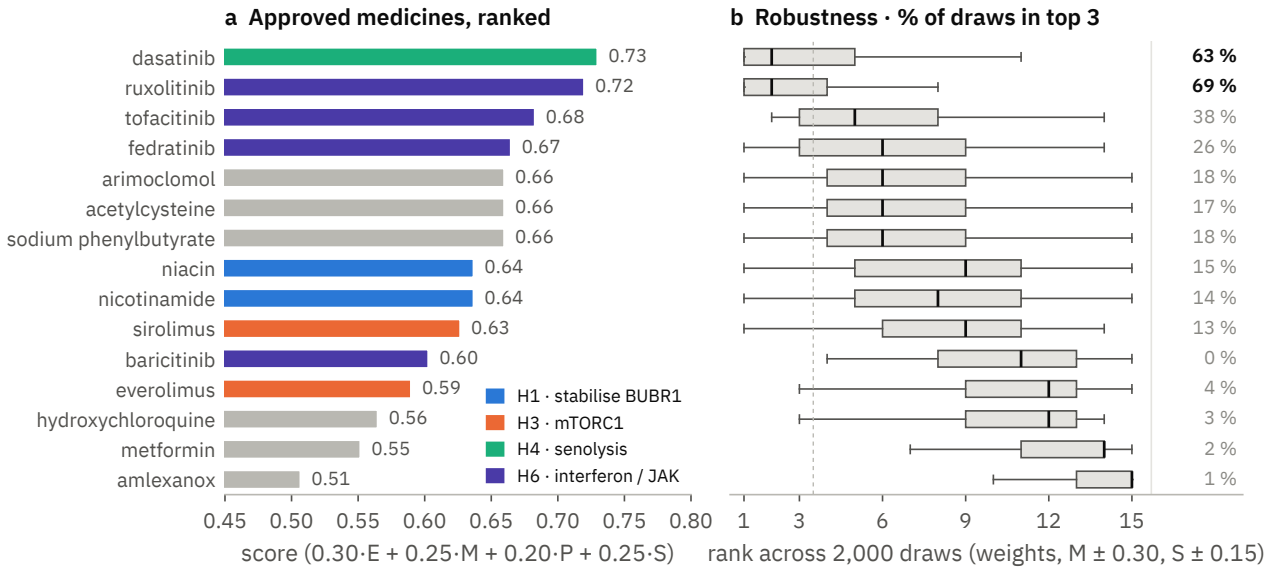


Figure 3. a, Score of the 15 approved medicines, coloured by therapeutic line; grey marks the lines this report does not propose (H2 closed, H5 not pursued, H7 declined). **b**, Rank distribution over 2,000 draws that perturb the weights (Dirichlet), each line's mechanistic fit (±0.30) and each line's safety (±0.15), with ties broken at random; right, share of draws in the top 3. Dasatinib (42 %) and ruxolitinib (38 %) share most first places; the other JAK inhibitors and the H7 agents follow, which is why the clinical tiering, not the score, carries the recommendation.

The score is an input to, not a substitute for, the clinical reasoning. Dasatinib leads, and tier 1 frames it as the one decisive, cheap experiment nobody has run rather than as a treatment. The JAK inhibitors score high on evidence because JAK inhibition has shown clinical benefit in human type I interferonopathies (a different disease, so graded *human_other*), but low on fit; ruxolitinib leads them because its oral label carries no boxed warning (unlike tofacitinib and baricitinib) and it has full paediatric feasibility (unlike fedratinib). They stay a biomarker-gated tier 4, because the interferon signature has never been

measured in any MVA patient. Sirolimus, first in an earlier draft, now ranks tenth (§4.4). The dasatinib + quercetin combination used in mouse studies includes a supplement, so we propose dasatinib as the approved agent and test both arms preclinically.

4.2 What each line does to an unstable cell, and what would change the answer

Ranking drugs by score answers "which is best?". It does not answer the question a clinician actually faces, which is what the drug *does* to a population of chromosomally unstable cells. We therefore label every line with one of three axes, declared in the knowledge base alongside the mechanism rather than assigned after seeing the ranking:

Axis	What it does	Lines
correct	fewer missegregation events: more functional BUBR1, so fewer errors are made	H1 (NAD^+/SIRT2), H2 (read-through), H7 (proteostasis)
clear	remove the cells that are already abnormal	H4 (senolysis), H5 (aneuploid-selective stress)
tolerate	leave the error rate alone and help the resulting cells cope	H3 (mTORC1), H6 (interferon/JAK)
avoid	do not add mitotic or genotoxic insult	S1 (safety card)

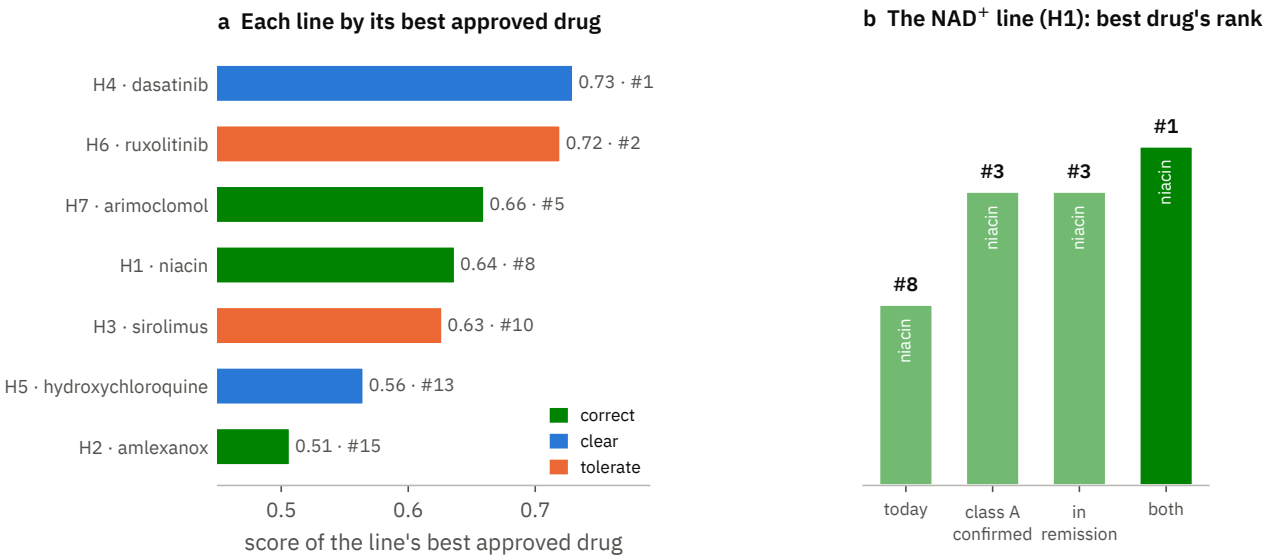


Figure 4. a, Each therapeutic line placed by the score of its best approved drug, coloured by axis. Summarised per line, not averaged over drugs: a mean over drugs would mostly report how many drugs we seeded per line. **b**, The rank of the corrective NAD^+ line's best drug under the two facts a clinical team can establish.

No axis wins on the evidence; the corrective lines are each blocked by a different, nameable fact about this child. That is the useful statement, and it is checkable line by line:

Line	Axis	Best drug	Rank	What blocks it
H4	clear	dasatinib	1	no senolytic ever tested in a BubR1 model; the best human senolytic RCT missed its primary endpoint
H6	tolerate	ruxolitinib	2	may weaken the immune clearance of aneuploid cells; no interferon signature has been measured in any MVA patient
H7	correct	arimoclomol	5	its mechanism *is* aneuploidy tolerance (§4.4)
H1	correct	niacin	8	allele class unmeasured; rhabdomyosarcoma depends on NAMPT/NAD^+
H3	tolerate	sirolimus	10	aneugenic in vitro; temsirolimus added myelotoxicity in paediatric rhabdomyosarcoma
H5	clear	hydroxychloro-quine	13	cell-line evidence only; would also deplete normal mosaic tissue
H2	correct	amlexanox	15	UGA-A stop context plus NMD; the compounds that correct UGA are unapproved

Two measurements would move a corrective drug to the top. Both concern H1 and both are things a clinical team can establish; nothing else in the model changes.

Scenario	Best overall	Second	Best corrective drug (rank)
As scored today	dasatinib	ruxolitinib	arimoclomol, declined (5)
Allele class A confirmed (rung-1 blot)	dasatinib	ruxolitinib	niacin (3)
Documented remission (NAMPT concern 0.35 → 0.10)	dasatinib	ruxolitinib	niacin (3)
Both	**niacin**	nicotinamide	**niacin (1)**

This is the opposite of a robustness claim, and we state it that way deliberately: the ranking *is* sensitive to these two inputs, which is exactly why the report's first recommendation is to measure them rather than to prescribe. An earlier draft instead claimed that six of the seven best-scoring medicines sat on the *tolerate* axis and that this was "not a weighting artefact". A blind adversarial review showed it was: the claim flipped when either of two curated H1 inputs was returned to its earlier value, and neither was inside the robustness analysis. We now perturb mechanistic fit over ±0.30 and safety over ±0.15, break ties at random, and publish the counterfactual table above instead of the claim.

The hazard the axis makes visible

In a cancer-predisposition syndrome, **a premalignant aneuploid clone is an abnormal cell too**. A drug on the *tolerate* axis cannot distinguish a struggling but healthy aneuploid cell in muscle from a clone on its way to a second malignancy; by construction, it helps both. That is explicit for H7, where the lever *is* an aneuploidy-tolerating mechanism (Donnelly et al. 2014), and it is why H3 and H6 are proposed only against named morbidity (muscle loss, inflammatory disease) with surveillance, never as chemoprevention. It is also why the safety note and the measurement plan, not a prescription, are what we put first.

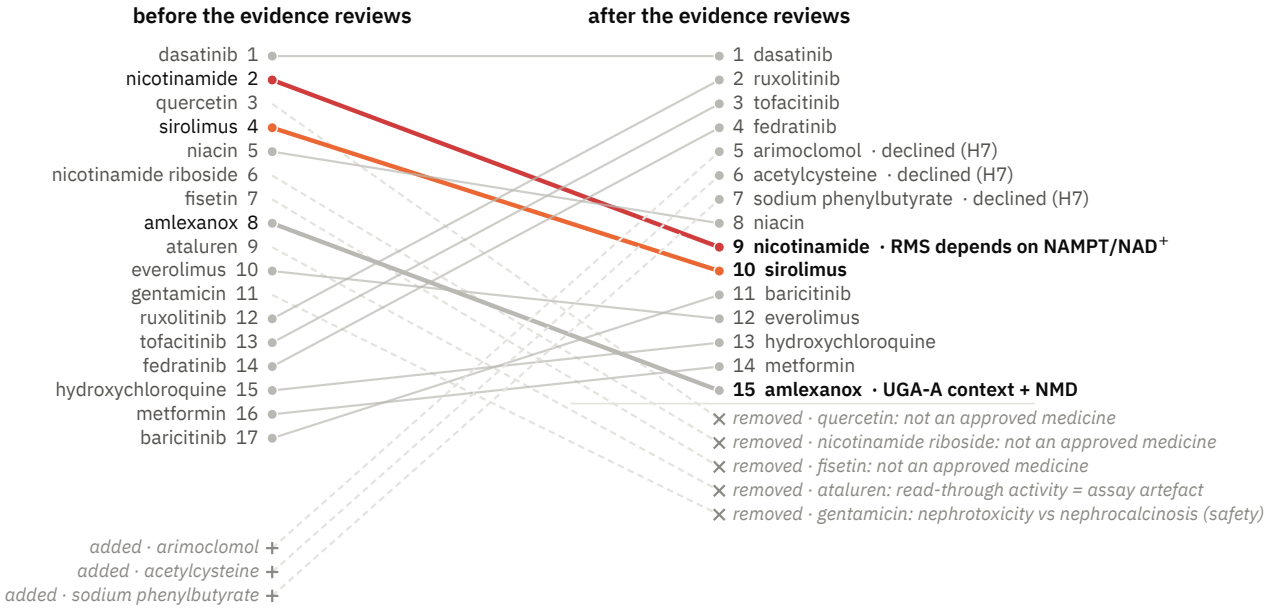


Figure 5. How the evidence reviews changed the ranking. Left, ranks before (literature + connectivity); right, after (approved medicines only). Nicotinamide fell because rhabdomyosarcoma depends on the pathway it feeds (§4.5); sirolimus rose on the first review and then fell again when its mouse evidence turned out to come from monoallelic carriers (§4.4); amlexanox fell when the read-through line was closed (§4.6); five products were removed for regulatory status, efficacy or safety, and the proteostasis line was added and declined.

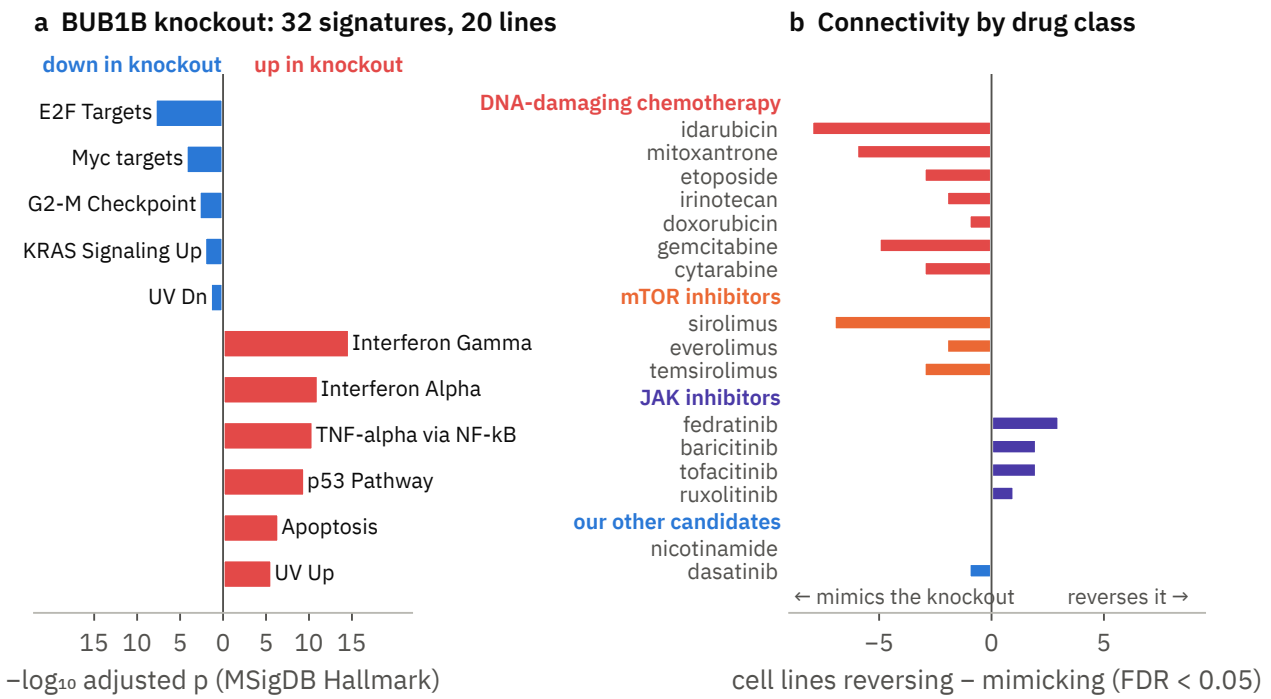


Figure 6. a, Hallmark pathways enriched among the 150 most up- and down-regulated genes of the consensus *BUB1B*-knockout signature (Enrichr). **b**, For selected drugs, number of cell lines in which the drug's signature significantly reverses the knockout signature minus the number in which it mimics it.

4.3 Connectivity: what losing BUB1B does to a cell, and which drugs push the same way

The signature. Interferon-γ and interferon-α responses, NF-κB and p53 dominate the up-regulated genes (top genes include *CDKN1A*, *ISG15*, *IRF7*); E2F, MYC and G2–M programmes dominate the down-regulated ones (Figure 6a). This is the expected response to chromosome missegregation: micronuclei sensed by cGAS–STING drive interferon signalling (Mackenzie et al. 2017; Bakhoum et al. 2018), p53 arrests the cycle, and proliferation shuts down (Santaguida et al. 2017).

Validity check: mimickers. The drugs that reproduce the signature in the most cell lines are dominated by DNA-damaging or anti-proliferative chemotherapies (idarubicin 8 lines, mitoxantrone 7, gemcitabine, etoposide, cytarabine, irinotecan), alongside other anti-proliferative agents such as sirolimus and bortezomib (8 lines each). A signature reproduced by genotoxic stress is biologically coherent, and it supports the safety card directly: in a child whose cells already sit in this state, these agents are predicted to add to it in normal tissue.

Reversers, and how we actually used them. 539 compounds reverse the signature significantly (FDR < 0.05) in at least one cell line, and 25 do so in at least three net lines, so a "top hit" here means little on its own. We did not pick H6 off the top of that list: the up-regulated arm is interferon-dominated, so we *looked for* JAK inhibitors in the reverser list and found four, three with no mimicking line and ruxolitinib with one, at ranks 15 (fedratinib), 38 (baricitinib), 43 (tofacitinib) and 154 (ruxolitinib). That is class consistency, not a data-driven discovery, and we report it as such. There is no permutation control for how often a four-member drug class achieves this by chance, which is the main reason H6 stays biomarker-gated at tier 4.

What the top of the reverser list actually contains is a better argument than the JAK inhibitors are. The 8th-ranked reverser is **tozasertib, an Aurora kinase inhibitor** — a drug predicted to make this disease worse, since it disables the same mitotic checkpoint machinery. The 24th is **EX-527, a SIRT1 inhibitor**, pointing against the very axis H1 is built on. Both reverse the knockout signature convincingly. This is the cleanest demonstration we can offer that **reversing a transcriptional signature is not a therapeutic claim**, and it is why connectivity sits beside the score rather than inside it.

Self-critique. mTOR inhibitors consistently **mimic** the signature (sirolimus 8 lines, everolimus 3, temsirolimus 3) because they deepen the anti-proliferative arm. We recorded this as contrary evidence for

H3, one of several reasons sirolimus now ranks tenth rather than first. The growth concern is partly allayed by 25 children on sirolimus without loss of linear growth (Hymes & Warshaw 2011), against one case report of growth failure (Rangel et al. 2009). *Limits:* the signatures come from tumour cell lines after acute knockout, not from patient cells with chronic partial deficiency; they are used for interpretation, safety and self-critique, not as evidence of efficacy.

4.4 Line-by-line rationale

S1 · Safety note (tier 0)

- *Mechanism, in both directions.* Vinca alkaloids, taxanes and eribulin kill through checkpoint-dependent mitotic arrest. With low BUBR1 the prediction is genuinely two-sided: cells may slip out of arrest and survive with more missegregation (less tumour kill, more chromosomal damage in normal dividing tissue), or fail to sustain arrest and die faster (more sensitivity, more toxicity). Checkpoint-weakened cells are a classical setting for antimitotic *hyper*sensitivity, so we present the ambiguity rather than resolve it. Either way the conclusion is closer monitoring, not avoidance.
- *Clinical signals in MVA.* Severe adverse reaction to dactinomycin in one biallelic *BUB1B* child (Ochiai et al. 2019); reduced-intensity chemotherapy for rhabdomyosarcoma in one PCS-MVA child (Nishitani-Isa et al. 2019); surveillance guidance for genomic-instability disorders (Nakano et al. 2024); poor growth-hormone response in four MVA patients, none of them *BUB1B* (§4.7).
- *Organ-specific.* BubR1 insufficiency causes cardiac hypertrophy and fibrosis in mice (PMID 40607964), which supports cardiac monitoring with anthracyclines. BubR1-hypomorphic mice also show *central* nervous system hypomyelination (PMID 27922816); peripheral nerves were not studied, so that paper does not by itself predict vincristine neuropathy, and we no longer claim it does.
- *The risk of the note itself is under-treatment.* This is the one deliverable most likely to be acted on, and the danger is that a flag against an antimitotic is read as a reason to de-escalate curative therapy. The note says so on its first line, recommends discussion and monitoring rather than substitution, and should be countersigned by a paediatric oncologist before use.
- *Deliverable.* A two-page pharmacogenomic note (`safety_card.pdf`): section A for drugs the child may actually receive, section B for the paediatric label items of every candidate ranked here (dasatinib pleural effusion and bone growth; sirolimus proteinuria, troughs, CYP3A4 and live vaccines; JAK inhibitors zoster, TB and live vaccines; fedratinib's boxed warning for Wernicke's encephalopathy).

H4 · Senescent cells (tier 1, as an experiment).

- *Evidence.* Genetic clearance of $p16^{Ink4a}$-positive cells delayed sarcopenia, cataract and lipodystrophy in BubR1-hypomorphic mice (Baker et al. 2011). It delayed onset and slowed progression; it did not rescue the phenotype and did not extend lifespan in that background.
- *Gap.* No senolytic drug has ever been tested in a BubR1 or aneuploidy model, which makes BubR1 H/L1002P mice (Sieben et al. 2020) the obvious and cheap experiment.
- *Contrary human evidence.* The best senolytic RCT (dasatinib + quercetin, 60 women) missed its primary endpoint (Farr et al. 2024); the pulmonary fibrosis pilot did not change lung function (Justice et al. 2019).
- *Connectivity, by the same standard as sirolimus.* Dasatinib reverses the knockout signature in 2 cell lines and mimics it in 3. We count mimicry against sirolimus, so it counts against dasatinib too: one more reason H4 is proposed as an experiment, not a treatment.
- *Caution.* Senescence has developmental roles, which argues for intermittent dosing and preclinical data first. Dasatinib has paediatric dosing and an oral suspension.

H3 · mTORC1 (tier 3, low-dose sirolimus; tenth by score).

- *What the mouse study actually shows.* In the BubR1 allelic series (Sieben et al. 2020), sarcopenia correlated with mTORC1 hyperactivity in mice carrying a **single** MVA allele. The biallelic genotype that mirrors

this child — truncating plus kinase-domain missense, X753/L1002P — **died prematurely**; the viable biallelic model is H/L1002P, a hypomorph plus missense. No mTOR inhibitor was given in that study. An earlier draft of this report attributed the mTORC1 result to "this child's allele class", which was wrong; the correction costs H3 its strongest-looking support and is the main reason sirolimus fell from first to tenth. The child is alive where that mouse genotype was not; a mouse-to-human comparison of survival is too unreliable to say more than that p.Asn1002Lys *might* retain more function than L1012P (§2.1).

- *Paediatric data, stated in full.* In ARST1431, temsirolimus added to VAC/VI in 297 children with intermediate-risk rhabdomyosarcoma gave **no event-free-survival benefit** (HR 0.86, 95 % CI 0.58–1.26, $p = 0.44$) and **more toxicity**: grade 3–4 anaemia in 58 % versus 41 %, and **one treatment-related death** in the temsirolimus arm (Gupta et al. 2024). This is simultaneously the best paediatric feasibility evidence for an mTOR inhibitor in exactly this tumour and a serious caution in a child with a weak checkpoint and marrow exposure.
- *Drosophila, labelled honestly.* Feeding rapamycin to a fly model of MVA-like microcephaly partially rescued **neuroblast number** (the survival of aneuploid neuroblasts) (González-Blanco et al. 2026) — by our own taxonomy a *tolerate* effect, not a correction — and did not rescue progeny number or brain size.
- *A safety concern that needs its own caveat.* Rapamycin caused chromosome malsegregation and CREST-positive micronuclei in human lymphocytes and lymphoblasts (Bonatti et al. 1998). It is a single 1998 in-vitro study, not replicated since, and its centrosome findings were in Chinese-hamster cells; set against it are decades of paediatric exposure in transplantation, tuberous sclerosis and vascular anomalies. We treat it as a reason to **monitor missegregation** (%MN-RET, §5) rather than as proof of harm.
- *If used.* Low, intermittent doses for muscle only, with phospho-S6, lean mass and growth endpoints, plus %MN-RET as a safety readout and the label items in section B of the safety note.

H1 · Raise BUBR1 abundance (tier 2, an NAD⁺ precursor, doubly gated).

- *Mechanism.* SIRT2 keeps BUBR1 Lys668 deacetylated, protecting it from degradation; falling NAD⁺ lowers BUBR1. SIRT2 overexpression extends the lifespan of BubR1-hypomorphic mice and the NAD⁺ precursor NMN raises BUBR1 in vivo (North et al. 2014). Lys668 lies upstream of codon 737, but the truncating allele's transcript is removed by NMD, so in practice the mark matters only on the missense allele's product.
- *Which precursor, and a problem with the obvious one.* The tested modality is NMN, which is not an approved medicine. Of the approved precursors, **nicotinamide is itself a non-competitive sirtuin inhibitor at physiological concentrations** (Bitterman et al. 2002) — it is the reaction product of the deacetylation this line depends on. **Niacin**, which enters via the Preiss–Handler route, carries no such objection and is the better mechanistic match; we score nicotinamide 0.2 lower on fit for this reason. That the two still finish level is a reminder that the score is a summary, not an argument.
- *Fit, and its condition.* The line exists only if the missense allele belongs to the low-abundance class, which is unmeasured but likely (prior 0.75, §2.1), giving $M = 0.8 \times 0.75 = 0.6$ before the per-drug adjustment. Nicotinamide has a large paediatric safety record (Gale et al. 2004; Knip et al. 2000).
- *Contrary evidence.* SIRT2 does not extend lifespan in wild-type mice (Wu et al. 2023); and, decisively for timing, rhabdomyosarcoma depends on NAMPT (§4.5).
- *Why this is tier 2 and not tier 4.* Both of its blockers are *measurable facts about this child*, not gaps in the literature: the allele class (one immunoblot) and remission status (already known to the treating team). Establish both and this line ranks first overall (§4.2) — which is why the report's main recommendation is a measurement.

H6 · Interferon / JAK–STAT (tier 4, biomarker-gated).

- *Rationale.* The up-regulated arm of the knockout signature is interferon-dominated; the four JAK inhibitors in LINCS reverse it as a class, though none near the top (§4.3); and JAK inhibition reduces the SASP and frailty in old mice (Xu et al. 2015).
- *Human target confirmation, and its limit.* A 2024 narrative review reports a chronic immune response in MVA patients (Malumbres 2024). **No interferon signature has been published for any MVA patient**:

the target is plausible, not confirmed, which is why this line is gated on measuring it in the child.

- *Paediatric precedent.* Baricitinib in children with interferonopathies, given under a compassionate-use protocol (NCT01724580) (Sanchez et al. 2018). Two later paediatric baricitinib or JAK-inhibitor studies in the same diseases, not part of that paper: NCT04517253 (Japan; terminated) and NCT03921554 (Aicardi–Goutières; completed).
- *Decisive caveat.* The inflammatory signal of aneuploid cells drives their immune clearance (Santaguida et al. 2017); blocking it could weaken tumour immunosurveillance. Not proposed for chemoprevention.

H5 · Aneuploid-selective stress (not pursued; one member is an anti-target here). AICAR, 17-AAG and chloroquine selectively impair aneuploid cells (Tang et al. 2011); metformin and hydroxychloroquine are approved proxies. The evidence is from cells only, and in a mosaic child these drugs could also deplete normal tissue. One of the three, however, turns out to be an **anti-target for this particular genotype**: 17-AAG acts on HSP90, and HSP90 is the chaperone that folds kinase-domain BUBR1 missense mutants — inhibiting it removes essentially all of the mutant protein while barely touching wild type, and blocking the proteasome prevents that loss (Suijkerbuijk et al. 2010). A drug chosen from the downstream phenotype would here destroy the residual protein the child depends on, if scenario A holds. The objection concerns 17-AAG, not the approved proxies, so it does not lower this line's evidence score; instead HSP90 inhibitors are added pre-emptively to the safety card.

H7 · Proteostasis amplification and antioxidants (considered, declined). This is a commonly proposed alternative for this genotype, so we evaluated it rather than passing over it in silence.

- *The lever is real.* Mutant BUBR1 stability depends on HSP90 and on escape from the proteasome (Suijkerbuijk et al. 2010), and aneuploid human cells are impaired in HSP90-dependent folding with a compromised heat-shock response, so raising HSF1 restores HSP90 expression and folding in exactly the cell state this child has (Donnelly et al. 2014). Arimoclomol amplifies that response and is an approved medicine with paediatric experience (Niemann-Pick C, with miglustat; Mengel et al. 2021); sodium phenylbutyrate is a chemical-chaperone comparator.
- *The objection comes from the same paper that validates the lever.* Donnelly et al. (2014) identify HSF1 overexpression as the **first described aneuploidy-*tolerating* event in human cells**: its effect is to make chromosomally abnormal cells viable. In a child with a cancer-predisposition syndrome and a history of rhabdomyosarcoma, a mechanism whose benefit is inseparable from the survival of chromosomally abnormal cells is on the wrong side of the trade-off. The same logic applies to the antioxidant arm, which accelerates tumour progression and metastasis in mouse models (Sayin et al. 2014; Wiel et al. 2019).
- *Where it would reopen.* Only if rung 1 confirms scenario A *and* an assay separates BUBR1 restoration from general survival of aneuploid cells — concretely, micronuclei must fall, not merely cell number rise. That is a falsifiable condition, not a polite refusal.

4.5 A conflict at the heart of our initial lead: NAD⁺ and rhabdomyosarcoma

What we found

Rhabdomyosarcoma, the tumour in this child's history (remission status to be documented), depends on **NAMPT**, the rate-limiting enzyme of the NAD⁺ salvage pathway. The clinical NAMPT inhibitor OT-82 depleted NAD⁺ and inhibited growth at ≤1 nmol/L in all ten molecularly diverse RMS cell lines, and produced complete tumour regression in orthotopic models (McKay-Corkum et al. 2023). Nicotinamide is the direct substrate of NAMPT; nicotinamide riboside bypasses NAMPT but still raises the NAD⁺ pool. NR supplementation also increased cancer prevalence and brain metastases in a breast-cancer model (Maric et al. 2023).

Against this, the ONTRAC phase 3 trial found that nicotinamide *reduced* new non-melanoma skin cancers by 23 % in 386 high-risk adults (Chen et al. 2015). That benefit is attributed to repair of UV-induced DNA damage, biology that does not transfer to a NAMPT-dependent sarcoma: ONTRAC supports tolerability, not oncological safety here. We **condition rather than delete** the line:

1. BUBR1 stabilisation is chronic and preventive; NAMPT dependency is an acute need of an existing tumour.
2. No NAD$^+$ precursor unless the child is in documented remission, with intensified surveillance (Nakano et al. 2024) as a stopping rule.
3. The first experiment must show that an NAD$^+$ precursor (niacin) raises BUBR1 in this child's own cells; if it does not, the risk is not worth taking.
4. On the safety card: if the tumour recurs, a NAMPT inhibitor would be a rational agent and a NAD$^+$ supplement would be expected to antagonise it.
5. Precedent for caution rather than prohibition: a completed n-of-1 study of NR in Li-Fraumeni syndrome, another cancer-predisposition syndrome (NCT03789175).

No selective SIRT2 activator exists (the published chemical matter is inhibitors), so the trade-off cannot currently be engineered away.

4.6 Closing the read-through line on allele-specific evidence

Read-through efficiency depends mainly on the stop codon and its +4 nucleotide, with the preference C > U > A/G (Mangkalaphiban et al. 2024). We computed the context of this allele from the public reference CDS (`contexto_stop.py`; Figure 7).

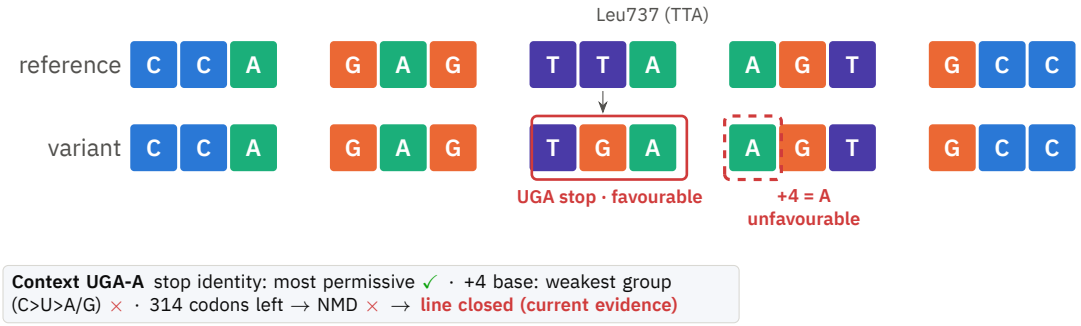


Figure 7. The allele-level argument. Reference and variant coding sequence around codon 737 (codons separated). The stop codon is the most read-through-permissive kind, but the base that follows it is in the least favourable group, and the transcript is expected to undergo nonsense-mediated decay.

This is a two-feature reading of the context (stop identity and +4 base); Mangkalaphiban et al. (2024) also publish a predictor trained on the extended context, which we did not run. Combined with drug-level evidence, it closes the line *on current evidence*:

- **Ataluren removed**: its read-through activity is attributed to an off-target luciferase artefact (McElroy et al. 2013); the cystic fibrosis phase 3 failed and EMA did not renew its conditional authorisation.
- **Amlexanox demoted**: its original report showed more nonsense-containing mRNA and full-length protein (PMID 22938201), but independent controlled comparisons did not reproduce the mRNA increase (PMID 33396210, PMID 37866515); its proposed action is NMD-related rather than read-through.
- **The compounds that correct UGA alleles are not approved**: 2,6-diaminopurine (Trzaska et al. 2020), G418 and SMG1 inhibitors, several of them toxic; the approved aminoglycoside gentamicin reads through only weakly and is nephrotoxic, a poor fit for a child with nephrocalcinosis.

The recommendation is an experiment, not a drug: a non-luciferase reporter carrying the native UGA-A context, with and without NMD inhibition, in patient-derived cells.

4.7 What has actually been given to people with MVA

We screened **every PubMed record on MVA or premature chromatid separation** (132 records; `revision_casos.py`) for treatment terms and read every hit. Only 9 mention any treatment, and none reports a disease-modifying drug.

Exposure	MVA type	What happened	Source
Dactinomycin for Wilms tumour	MVA1 (*BUB1B*)	severe adverse reaction prevented further chemotherapy	PMID 31081598
Chemotherapy for rhabdomyosarcoma	PCS-MVA	reduced-intensity chemotherapy	PMID 31184400
Chemoradiotherapy *advised*, orbital embryonal RMS after Wilms	MVA3 (*TRIP13*)	second malignancy; treatment recommended, but neither its delivery nor any outcome is reported	PMID 42595739
Stem cell transplant (reduced-intensity conditioning) for MDS with monosomy 7	MVA1 (*BUB1B*)	tolerated, haematological recovery, then graft rejection and fatal sepsis	PMID 31053147
Growth hormone for short stature	MVA2 (*CEP57*) ×3, MVA ×1	poor or no response in all four; possible growth-plate resistance	PMID 35434947 and 3 others

Two conclusions feed the safety card. Growth hormone combines low expected benefit with a cancer-predisposition concern. Cytotoxic therapy is poorly tolerated in MVA1, and reduced-intensity regimens are already the published practice.

5 How we would know whether it works in this child

Every repurposing report, including the earlier drafts of this one, ends at "it should be tested". For a disease with fewer than 50 known patients that is where the real problem starts, because the usual endpoints do not work: tumour incidence needs decades and a cohort, and a lymphocyte karyotype reports an aneuploidy burden accumulated over years, so it cannot tell whether a drug started last month is helping. What is missing is an endpoint that moves in weeks.

The endpoint: micronucleated reticulocytes (%MN-RET)

What it counts	micronuclei — fragments left outside the nucleus by a missegregation event. It is the direct product of this disease's defect, not a proxy for it
Why reticulocytes	they circulate for 1–2 days, so the measurement is a **recent error rate**, not a lifetime accumulation. The spleen removes micronucleated erythrocytes, which is why the mature red-cell assay fails in humans and the reticulocyte one works
How	one blood draw; flow cytometry scores ≥20,000 $CD71^+$ reticulocytes per sample (Dertinger et al. 2003; Dertinger et al. 2011)
Paediatric reference	upper bound of the 90 % tolerance interval in paediatric controls = **0.42 %** (Baig et al. 2020)
Status in MVA	**never measured**. This is rung 0, and it is worth doing whether or not any drug is ever given: the first result is diagnostic in the sense of a disease-activity observation (not yet a validated diagnostic test) — whether this disease is visibly active in blood, and how much — and it applies to every MVA patient, not only this one
Two-way information	it falls if a drug reduces missegregation and **rises if a drug adds it** — which is why it gates sirolimus, once our top-scoring candidate (§4.4)

The design, and what it can actually resolve. We simulated an n-of-1 trial on this endpoint (`scripts/t2/nof1.py`; public parameters only). Alternating on/off blocks; with 20,000 $CD71^+$ reticulocytes per sample the expected count is in the hundreds, so $\log(y/N)$ is near-normal. **The between-visit variability σ is not assumed known** — it is estimated from the same data, so the posterior for the log rate ratio is a Student t with $n-2$ degrees of freedom. An earlier version conditioned on the true σ, which inflated power and understated false positives for a quantity the report itself says has never been measured. Every cell uses common random numbers and carries its Monte Carlo standard error.

Three results, and the first is unwelcome:

1. **The strict rule is too strict to be useful.** Requiring $P(\text{reduction} \geq 25\,\%) \geq 95\,\%$ detects only a **63 % or larger** true reduction at 80 % power with four visits per condition; a 35 % reduction is caught 12 % of the time. Its false-positive rate is under 1 %, far below the conventional 5 %, which is the price being paid. The conventional rule — any reduction with 95 % posterior certainty — is calibrated at the advertised rate and needs a 51 % reduction, or 9 visits per condition to reach a 35 % one. Proposing this endpoint without the power calculation would be proposing an experiment that cannot answer its own question.

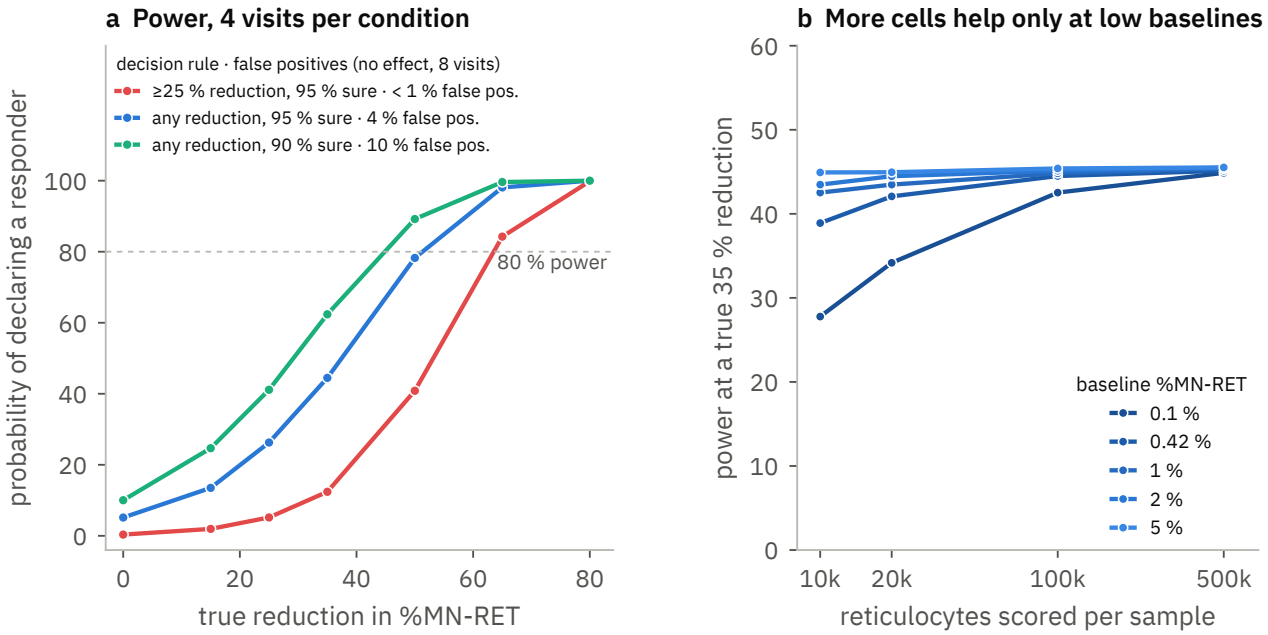


Figure 8. a, Power against the true reduction in %MN-RET for three candidate responder rules, with four on/off visits per condition, $\sigma = 0.35$, 20,000 reticulocytes per sample and an assumed 2 % baseline; each rule is labelled with its own false-positive rate. **b**, Power at a true 35 % reduction by baseline and by cells scored: counting more cells buys almost nothing except at baselines near or below the paediatric reference.

2. **Counting more cells helps only if the baseline is low.** At the assumed 2 % baseline, going from 10,000 to 500,000 reticulocytes per sample moves power from 43.5 % to 45.5 %, because the counting term $1/y$ is negligible beside $\sigma^2 = 0.12$. At a baseline of 0.1 % — below the paediatric reference ceiling — the same change moves it from 27.8 % to 44.9 %. So the rule is conditional, and rung 0 establishes which regime applies: *if the baseline exceeds about 1 %, spend the budget on repeated visits rather than deeper runs.*
3. **The decision rule dominates every other knob.** Power at a 35 % reduction ranges from 12 % under the strict rule to 62 % under "any reduction, 90 % certainty", whose false-positive rate is 10 %. That trade-off — not the power figure alone — is what a clinical team and a family have to choose between, and the right time to choose is before the first sample.

Every number above is conditional on σ, the between-visit variability, which has never been measured in anyone with MVA. **So the first study is not a drug study**: establish the baseline and σ in this child with repeated sampling off treatment. That single measurement also tests the disease model — a %MN-RET within the paediatric reference range would be evidence against the disease model on which the measurement-led part of this report rests, and we would rather find that out from a blood draw than from a failed treatment. **A ready-to-use one-page protocol** for that study (`reports/t2/rung0_protocol.pdf`) specifies the samples, the confounders to record (infection and malnutrition raise %MN-RET in children; cytotoxic therapy raises it up to 26-fold, peaking 3–5 days after starting), the pre-declared decision table, and how many on/off visits each responder rule needs for the σ that the child turns out to have.

6 How a medicine like this gets validated

No clinical trial for MVA is registered anywhere (ClinicalTrials.gov, October 2026). With fewer than 50 known patients a randomised trial is impossible, but comparable diseases have walked a route with four rungs (Figure 9).

Making one patient's treatment into evidence. Define the responder in advance (a common weakness of published n-of-1 trials); use the patient as their own control with on/off blocks and Bayesian analysis for reversible endpoints; and use natural history as an external control for endpoints that cannot be switched off. Progeria shows this works: lonafarnib was approved on a single-arm cohort compared with untreated natural-history controls, with mortality 3.7 % vs 33.3 % (Gordon et al. 2018). The closest

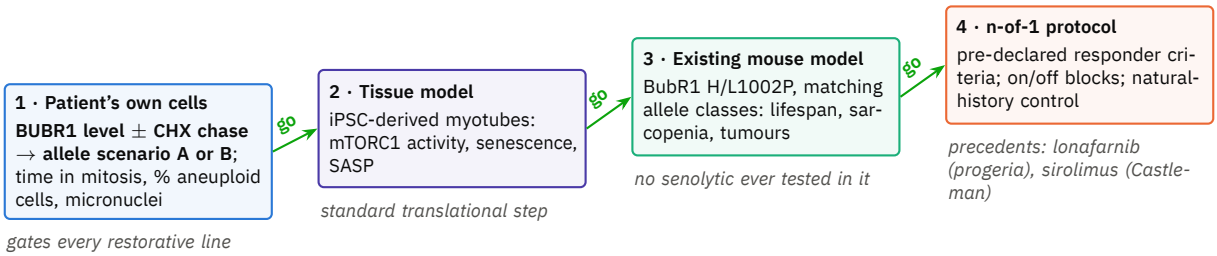


Figure 9. The validation ladder. Each rung has a go / no-go readout. Preceding all of them is **rung 0** (§5): measure %MN-RET and its between-visit variability in this child, off treatment — it costs a blood draw, it is the only endpoint that moves in weeks, and it can falsify the disease model before anyone is exposed to anything. Rung 1 then decides the allele scenario, and with it whether any restorative line exists.

precedent to this project is **sirolimus in idiopathic multicentric Castleman disease** (Fajgenbaum et al. 2019): a pathway measured in the patients themselves, an approved drug repurposed against it, remission, then a prospective trial (NCT03933904). Our pipeline applies the same logic one step earlier.

Regulatory routes (checked against primary texts; the patient's jurisdiction is not known to us).

Situation	United States	European Union
Physician prescribes an approved drug off-label	outside FDA's IND rules: "this part does not apply to the use in the practice of medicine for an unlabeled indication" (21 CFR 312.2(d))	regulated nationally; typically the physician decides, documents the justification, obtains informed consent and follows the centre's protocol, usually with no case-by-case agency authorisation (in the EU, under each member state's rules on medicines in special situations)
Structured n-of-1 study of an approved drug	may be IND-exempt if 21 CFR 312.2(b)(1)(i)–(v) hold (no labelling claim, IRB, consent); condition (iii) on risk in a special population needs FDA discussion	can qualify as a **low-intervention clinical trial** (Reg. (EU) 536/2014, art. 2(2)(3)): authorised medicine used in an evidence-based way, minimal added risk
Expanded access / compassionate use	Form FDA 3926; for *investigational* drugs only	Art. 83 Reg. (EC) 726/2004; for *unauthorised* medicines only

In both frameworks the barrier to trying a ranked candidate is clinical and ethical, not regulatory, and both expect documented target engagement, which the first rung supplies. Sirolimus, everolimus, dasatinib and tofacitinib all exist in oral liquid or dispersible forms that can be dosed by weight in a child.

What we would do first: (0) %MN-RET and its variability in this child, off treatment, which costs a blood draw and sets up everything else; (1) patient cells, to decide the allele scenario and so gate H1 and H7; (2) BubR1 H/L1002P mice with three arms (vehicle, dasatinib, dasatinib + quercetin), as the gate for H4; (3) a non-luciferase read-through reporter, as the reopening condition for H2; (4) only then, and only in remission, an n-of-1 protocol with the clinical team, with the responder rule agreed in advance.

7 Scalability

- **New patient or gene:** a new `hipotesis.yaml`; `cmap.py`, `revision_casos.py`, `contexto_stop.py` and `estructura.py` all take a gene or a change as a parameter.
- **Tested, not just claimed:** re-running connectivity for the other MVA and checkpoint genes (`cmap.py --comparar`) showed that CEP57, TRIP13, BUB3 and MAD2L1 have **no knockout signatures in LINCS**, so for them the pipeline runs on literature and labels only. BUB1 and TTK are profiled; their signatures correlate weakly with BUB1B's (Spearman ρ 0.06 and 0.27), sharing the proliferation and p53 response but not the leading interferon programme. H6 is therefore supported for BUB1B specifically, and the pipeline reports where a data source does not exist instead of filling the gap.
- **Staying current and cheap:** ChEMBL target expansion can add newly approved drugs automatically (in this run it added none; see the limitations); label mining keeps safety current; all sources are public and a full run takes minutes on a laptop.

8 Limitations

- Efficacy evidence is preclinical; no drug has been tested in MVA patients.
- **The class of the missense allele is unmeasured** (§2.1). The 0.75/0.25 prior is a judgement calibrated on one study, and under scenario B two of our seven lines do not exist.
- **The ranking is sensitive to two curated inputs for H1**, and we publish the counterfactual rather than claim robustness (§4.2). More generally, every mechanistic-fit value is an expert judgement; they are exposed in the YAML and stress-tested over ± 0.30, with safety over ± 0.15.
- **The evidence term still depends on curation depth.** Contrary studies are now weighted by level and capped, and harms moved to safety, but a line we interrogated harder can still score lower than one we did not. The per-line table in `candidatos.md` reports how many items of each kind we curated, so the asymmetry is visible.
- **Every ranked drug was hand-seeded.** The automated ChEMBL target expansion contributed no new drug here — mechanism annotation for spindle-checkpoint biology is too sparse — so the composition of the candidate list is ours, which is why axis conclusions are reported per line rather than averaged over drugs.
- **The power calculation rests on an assumed %MN-RET baseline and variability** that have never been measured in MVA. σ is now estimated from the data rather than assumed known, and every figure carries its Monte Carlo error, but the absolute numbers are conditional on the baseline rung 0 will establish.
- The three therapeutic axes are a deliberate simplification: a real drug can act on more than one, which is exactly why H7 is recorded with a primary and a risk axis rather than a single label.
- Connectivity signatures come from tumour cell lines after acute knockout, and we have no permutation control for how often a drug class reverses the signature by chance (§4.3).
- Label mining is keyword-based and US-centric; flags were reviewed manually. No dosing, drug–drug interaction or formulation-stability analysis was performed.
- The case-literature screen covers titles and abstracts (122 of 132 records have one), so details in full texts may be missed.
- Senescent-cell burden has never been measured in MVA patients.
- Regulatory routes are summarised for orientation, not as legal advice.

9 Data protection and LLM use

- **LLM declaration** (organisers' format, discussion #2): *Anthropic Claude via the Claude Code CLI, Claude Max (consumer tier), model-training opt-out enabled ("Help improve Claude" off), no feedback provided on any output, no content flagged for safety review.* Opted-out conversations are not used for training and are retained for 30 days.
- LLM sub-agents assisted the literature screening. **Every citation was then verified against PubMed, ClinicalTrials.gov or the primary regulatory text**; one misattributed first author was corrected, and claims that could not be verified are not cited.
- Track 2 used only gene-, pathway- and drug-level information. The one sequence-level analysis (§4.6) used the public reference CDS and the already-submitted coding change.

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Data: LINCS L1000 via SigCom LINCS; ChEMBL; openFDA (labels, NDC); ClinicalTrials.gov; Enrichr (MSigDB Hallmark 2020); AlphaFold DB (AF-O60566-F1 v6); UniProt O60566; Ensembl.

A Methods description form (Track 2 methods)

Answers to the organisers' official methods description template (version with the required LLM question), reproduced here because the submission form has no separate upload field for it.

Team name

aperezcastro (Agustín Pérez Castro, individual)

Please describe your approach in detail (how you went from variant/mechanism to candidate medication(s)).

Allele-aware, patient-aware pipeline (scripts/t2/). (1) Mechanism model from Track 1: null allele p.(Leu737*) (UGA, NMD) + kinase-domain missense p.(Asn1002Lys) → partial loss of function in a dosage-sensitive checkpoint. The CLASS of the missense allele (low abundance vs stable-but-dysfunctional) is unmeasured; every disease-associated substitution tested in or near the kinase domain was 5-10x less abundant and the known stable alleles lie outside that domain, so we carry both scenarios at a 0.75/0.25 prior and price the uncertainty into mechanistic fit, not evidence. AlphaFold structure: Asn1002 buried high-confidence residue near the MVA1 allele L1012P (destabilisation). (2) Knowledge base (hipotesis.yaml): 7 therapeutic lines + a safety card, each with PMID-cited evidence graded by level and direction (supporting/contrary): H1 stabilise BUBR1 (NAD+/SIRT2), H2 read-through/NMD inhibition of the nonsense allele, H3 mTORC1 (sarcopenia), H4 senolytics, H5 aneuploid-selective stress, H6 interferon/JAK-STAT (biomarker-gated), H7 proteostasis amplification/antioxidants (evaluated and declined). (3) Connectivity (cmap.py): consensus of 32 BUB1B CRISPR-knockout L1000 signatures (20 cell lines) via SigCom LINCS, two-sided rank enrichment against chemical perturbation signatures, cross-cell-line consistency, Enrichr interpretation (interferon/NF-kB/p53 up, E2F/G2M down). Used to check H6 for class consistency (JAK inhibitors reverse the signature), to support the safety card (DNA-damaging drugs mimic it) and to challenge H3 (mTOR inhibitors mimic it). (4) candidatos.py: seeded approved drugs (ChEMBL target expansion is available but added none in this run), annotated with ChEMBL and openFDA labels (paediatric use, boxed warnings) and penalised by the patient's HPO-derived risk profile; transparent score 0.30 evidence + 0.25 mechanistic fit + 0.20 paediatric feasibility + 0.25 patient-specific safety, with Dirichlet weight sensitivity; only approved medicines are ranked (discussion #5). (5) Clinical tiering by benefit-risk: tier 0 safety note plus rung-0 measurement; tier 1 intermittent dasatinib (highest-scoring approved drug), to be tested first in the BubR1 H/L1002P mouse; tier 2 an NAD+ precursor - niacin (nicotinamide ties on score but inhibits sirtuins) - gated on the allele class AND documented remission; tier 3 low-dose sirolimus for muscle with %MN-RET as a safety readout; tier 4 JAK inhibitors only if an interferon signature is measured; read-through closed on current allele-level evidence; proteostasis declined because raising HSF1 is a documented aneuploidy-TOLERATING event (Donnelly 2014). (6) nof1.py: an n-of-1 power simulation on micronucleated reticulocytes (%MN-RET), the only endpoint for this disease that moves in weeks. Validation ladder: rung 0 measure %MN-RET and its variability; patient cells (decides the allele scenario); iPSC myotubes; the BubR1 allelic mouse; then an n-of-1 protocol with pre-declared responder criteria.

(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."

Anthropic Claude via the Claude Code CLI (Claude Opus models), Claude Max (consumer tier), model-training opt-out enabled ('Help improve Claude' off), no feedback provided on any output, no content flagged for safety review - per the organisers' confirmed guidance of 28 Sep 2026 (discussion #2). Applicable terms: opted-out conversations are not used for training and are retained 30 days. Additional data minimisation: the model only saw code, aggregate counts, HPO terms and short candidate lists ($\leq$ 50 annotated variants); raw data and genome-scale derivatives were blocked from the agent by configuration. No patient variant was sent to any third-party API.

Was candidate identification automated (e.g., via a drug-target database query or LLM-assisted search), or primarily manual literature review and expert curation?

Hybrid: candidate expansion, annotation, safety flags, connectivity and scoring are automated; the hypotheses, evidence grading, mechanistic-fit values and final clinical tiering were curated manually (LLM-assisted literature search with every reference verified against PubMed).

Please describe any manual literature review or expert curation in detail.
PubMed searches at gene/pathway/drug level only (no patient data). Each claim in hipotesis.yaml cites a PMID whose title/abstract was retrieved and checked; contrary evidence is recorded explicitly (e.g., SIRT2 does not extend lifespan in wild-type mice; aneuploid-cell immune clearance argues against JAK inhibition). Mechanistic-fit values (0-1) are justified in the YAML. Automated label flags were reviewed manually.

Please state whether you relied solely on publicly available drug/target databases and literature (preferred), or whether proprietary data sources were also used.
Solely public databases and literature.

Please describe the public data sources used in detail (e.g., DrugBank, ChEMBL, ClinicalTrials.gov, OMIM, PubMed literature search).
PubMed/NCBI E-utilities; ChEMBL REST API (molecules, mechanisms, targets); openFDA drug labels; ClinicalTrials.gov API v2; LINCS L1000 via SigCom LINCS (l1000_xpr knockout and l1000_cp chemical signatures); Enrichr (MSigDB Hallmark 2020); UniProt O60566; AlphaFold DB AF-O60566-F1 v6; OMIM/HPO.

Please describe any proprietary data sources used in detail.
None.

How did you characterize the variant's mechanism (e.g., loss-of-function vs. gain-of-function, pathway disrupted, downstream biological consequence)?
Loss of function by reduced dosage: the nonsense allele is predicted to undergo NMD (no protein). For the kinase-domain missense we DO NOT assert a class: Suijkerbuijk 2010 splits MVA alleles into low-abundance and functionally-impaired classes, and although all disease-associated substitutions tested in or near the kinase domain were 5-10x less abundant, this residue has never been characterised, so scenario A (0.75) and scenario B (0.25) are both carried and a BUBR1 immunoblot in patient cells is the pre-declared gate. Pathway: spindle assembly checkpoint / mitotic checkpoint complex (APC/C-CDC20 inhibition) and kinetochore-microtubule attachment. Downstream: chromosome missegregation → micronuclei → cGAS-STING interferon/NF-kB and p53 responses (confirmed in the knockout signature), senescence/SASP, mTORC1 hyperactivity in muscle (mouse allelic series), tumour predisposition.

Please provide an estimate of time or effort spent on this analysis (if possible).
About 2 working days of analysis and writing; computation runs in minutes on a laptop (public APIs, cached).

Provide a method abstract (up to 500 words). Please include strengths and limitations of your approach as well as methodological detail.
We present an allele-aware, patient-aware repurposing pipeline for a child with biallelic BUB1B mosaic variegated aneuploidy (null p.Leu737* + kinase-domain p.Asn1002Lys). The mechanism is partial loss of function in a dosage-sensitive checkpoint; the missense allele class is unmeasured and both scenarios are carried. Three evidence streams were combined: (i) a PMID-cited knowledge base of seven therapeutic lines plus a safety card, scoring supporting and contrary evidence explicitly and weighting target validation below demonstrated benefit; (ii) connectivity analysis of 32 BUB1B-knockout L1000 signatures across 20 cell lines; (iii) automated annotation from ChEMBL and FDA labels, penalised by the patient's HPO-derived risk profile, with Dirichlet weight-sensitivity analysis. The knockout signature is interferon/NF-kB/p53-up and E2F/G2M-down; DNA-damaging chemotherapeutics reproduce it and mTOR inhibitors mimic it; JAK inhibitors reverse it with class consistency but at ranks 15-154 of 539 reversers, a targeted search rather than a discovery. Two evidence reviews changed most lines: read-through was closed, on current evidence, by the stop-codon context (UGA, unfavourable +4 A, NMD) plus evidence that ataluren's activity is a luciferase artefact; nicotinamide fell because rhabdomyosarcoma depends on NAMPT/NAD+ and nicotinamide inhibits sirtuins; senolysis became a preclinical experiment after the best senolytic RCT missed its endpoint; and sirolimus fell from first to tenth when its mouse support proved to come from monoallelic carriers. THE MAIN RESULT is a decision rule: classifying every line by what it does to a chromosomally unstable cell (correct fewer errors / clear the abnormal cells / tolerate help them cope) shows that no axis wins on evidence and that each corrective line is blocked by a different nameable fact about this child. Today the best-supported option is intermittent dasatinib (clear axis, narrowly ahead of the JAK inhibitor ruxolitinib); the mechanism-matched candidate, an

NAD+ precursor (niacin), ranks eighth, and rises to first if the clinical team establishes the allele class and documented remission - so the first recommendation is a measurement, not a prescription. In a cancer-predisposition syndrome a premalignant clone is an abnormal cell too. Deliverables: a two-page pharmacogenomic safety card, usable now; %MN-RET, a blood measurement never made in MVA, diagnostic first and then an endpoint with its power calculation; a ranked and conditioned candidate list, and a four-rung validation ladder (patient cells, iPSC myotubes, the published BubR1 allelic mouse model, then an n-of-1 protocol with pre-declared responder criteria) built on verified precedents (lonafarnib, sirolimus in Castleman disease). Strengths: per-allele strategies, explicit contrary evidence that demoted our own candidates, patient-specific safety beyond drug labels, and reproducible gene-agnostic code. Limitations: efficacy evidence is preclinical; connectivity signatures come from tumour cell lines after acute knockout; mechanistic-fit values are expert judgements and the ranking is sensitive to two of them, so the counterfactual is published rather than a robustness claim; every ranked drug was hand-seeded (automated target expansion added none); the %MN-RET baseline and variability used in the power calculation have never been measured in MVA; regulatory routes are orientation, not legal advice. An independent blind adversarial review of an earlier draft drove these corrections.